

EXERCISE 1: JOB-ORDER COSTING

Woodie Corp. builds customized carpentry and received a job for 2 chairs.

Per unit	Table	Chair
Wood	4 kg	1 kg
Senior carpenter	1 hour	1 hour
Junior carpenter	2.5 hours	2 hours

Information from purchasing and human resource departments are as follows.

Wood, \$ per kg	\$20
Senior carpenter, \$ per hour	\$100
Junior carpenter, \$ per hour	\$25

POHR for total manufacturing overheads is also applied at \$10 per Junior carpenter hours.

Determine the total cost of the job.

EXERCISE 2: APPLIED MANUFACTURING OVERHEADS

For June, Ace Inc. had the following budgets.

- For materials, Ace budgeted \$50,300 of direct materials and \$27,300 of indirect materials used in production.
- Direct labour is budgeted at \$150,150, while indirect labour for production is \$40,950.
- Other manufacturing overheads was estimated to be \$77,350.

The firm budgeted a consumption of 9,100 direct labour hours, and 45,500 machine hours to produce 40,000 units.

In reality, Ace Inc only used 8,500 direct labour hours, 42,000 machine hours to produce 37,500 units. Ace also incurs the following actual costs incurred for June:

- Direct materials of \$40,400 and indirect materials used in production of \$25,200.
- Direct labour of \$140,250 and indirect labour for production is \$36,600.
- Other manufacturing overheads of \$70,200

Ace Inc. uses normal costing. Total manufacturing overhead is allocated based on machine hours. Ace Inc. uses the direct method to write off over-/under-applied overhead to COGS.

Determine the over/under-applied MOH for June. What will be the impact on COGS?

EXERCISE 3: JOB-ORDER AND APPLIED MANUFACTURING OVERHEADS

Woody Corp. builds customized carpentry. For March, the financials were as follows.

	Actual (\$)	Budget (\$)
Direct materials	600	500
Direct labor @ \$15 per hour	1,050	1,125
Utilities	270	300
Workshop rental	800	800
Misc. manufacturing expenses	380	400

At the beginning of March, Woody Corp had 1 outstanding job (Job #10). In March, Woody Corp finished Job #10 and began and finished Job #11. Both were delivered to the customers before the month ended.

	Job #10 (\$)	Job #11 (\$)
Work done before March:		
Direct materials	\$100	
Direct labour @ \$15 per hour	\$300	
Applied MOH	\$400	
Work done during March:		
Direct materials	\$250	\$350
Direct labour hours	28 DLH	42 DLH

The firm uses normal costing. Total manufacturing overhead is allocated based on direct labor hours.

1. What is the total manufacturing overheads applied during March to Job #10 and Job #11 respectively?

2. Determine the total cost of Job #10 and Job #11

3. Determine the following balances for Woody Corp.
 - a. Work-in- process, as at 1 March
 - b. Work-in-process, as at 31 March
 - c. Finished goods, as at 31 March

4. Woody Corp uses the direct method to write off over-/under-applied overhead to COGS. Determine the COGS for the month of March.

EXERCISE 4: PREDETERMINED OVERHEAD RATES

A manufacturing startup has just built a new factory that can run 20,000 machine hours at full capacity.

As a startup, the firm believes it will be running at 60% of its full capacity next year. Other budgeted figures for the year are as follows:

Units produced and sold	4,000
Direct materials, per unit	\$3
Direct labor, per unit	\$4
Manufacturing overheads	\$42,000
Selling expenses	\$30,000

Manufacturing overheads are driven and allocated based on machine hours.

Calculate the **budgeted product cost** when the pre-determined manufacturing overhead rate is based on

- a) Estimated activity
- b) Capacity